Course Title	µP Interfacing and Programming	**Course Code**	EE3012
Department	Department of Electrical Engineering (DEE)	**Campus**	Lahore
Knowledge Profile	Engineering Practice (WK6)	**Credit Hrs.**	3+1
Knowledge Area	Computer Engineering (KA01)	**Grading Scheme**	Relative
HEC Knowledge Area	Depth Electives	**Applicable From**	Fall 2025
SDG	9 | Industry, Innovation and Infrastructure	**PBL**	1
Pre-requisite(s)	EE/EL1005 Digital Logic Design		

Course Objective	The course introduces the architecture, programming, and interfacing of microprocessor/microcontroller-based systems. A portion of this course comprises of assembly/C language programming to understand the relation of software and hardware in microprocessor/microcontroller-based systems. At the end of the course the student should be able to utilize a set of concepts common to design and implement stand-alone microcontroller-based systems which includes design, analysis of design, development of software and hardware required, and testing of the system using different available tools.

No.	Assigned Program Learning Outcome (PLO)
1	An ability to apply knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.
3	An ability to design solutions for complex engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations

A = Assignment, Q = Quiz, M = Midterm, F=Final
.

No.	Course Learning Outcome (CLO) Statements	Assessment Tools	Taxonomy Levels	PLO
1	**Describe** the internal architecture, programming of microprocessor and microcontroller.	A1, Q1, M1	C2	1
2	**Develop** a program to interface various peripherals with pins and ports of microcontroller.	A2, Q2, M2	C6	3
3	**Construct** a program to interface a microcontroller with external and internal components.	Q3, M2, F	C6	3
4	**Design** a system using micro-processor and micro-controller.	Q4,A3, F	C6	3

Text Books	**Title**	PIC Microcontroller and embedded systems (2nd Edition)
	Author	Muhammad Ali Mazidi
	Publisher	Pearson; 2nd edition
	Title	The PIC microcontrollers
	Author	Nebojsa Matic
	Publisher	mikroElektronika, 2003 (Free Available Online)
Reference Books	**Title**	Microprocessors and Interfacing
	Author	Douglas V. Hall
	Publisher	McGraw-Hill, 2006, 2nd Edition, ISBN 0-07-060167-4

Week	**Course Contents/Topics**	**Chapter**	**CLO**
01	Introduction to computer architecture, microprocessor 8086 and Addressing Modes	1, 2 (TB-1)	1
02	Software model, Instruction Set (Data Transfer instructions 8086)	3,4 (TB-1)	1,2
03	8086 Instruction set (Arithmetic and Logic instructions)	3,4 (TB-1)	1,2
04	x86 Program flow Instructions, I/O and Hardware Specifications	5,6,8 (TB-1)	1
05-06	General Microcontroller Architecture and PIC Microcontroller Architecture	1, 2 (TB-2)	1
07	Instruction Set (Data transfer, Arithmetic) PIC microcontroller	3 (TB-2)	2
08	Instruction Set (Logic, Shift Rotate instructions) PIC microcontroller	3 (TB-2)	2
09	Instruction Set (Program Flow instructions) PIC microcontroller	3 (TB-2)	2,3
10	I/O configuration and Interfacing (PORTS of PIC microcontroller)	4,6 (TB-2)	2,3
11	Timers and Control units of Controller (PIC microcontroller)	4,6 (TB-2)	3,4
12 – 13	Interrupts (External, internal, and other interrupts of PIC microcontroller)	4,6 (TB-2)	3,4
14 – 15	Reading and writing to EEPROM (PIC microcontroller)	4,6 (TB-2)	2,3
16	Writing mix Assembly and C code and Project Building using PIC microcontroller	4,6 (TB-2)	4

***TB-1 stands for Text book (The Intel Microprocessors)**

Assessment Tools	**Weightage**
Quizzes, Assignments*	20%
Midterm (I+II)	15+15 = 30%
Final Exam	50%

*Assignments/Quiz will also comprise of a final hardware based project